



SEMESTER 1 EXAMINATIONS 2025/2026

MODULE: CSC1047 - Advanced Algorithms & A.I. Search

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,3

EXAMINER(S):

Mark Humphrys	(Internal)	(Ext:8059)
Hemant Kumar Singh	(Internal)	(Ext:)

TIME ALLOWED: 2 Hours

INSTRUCTIONS: See inside.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

Multiple choice

Answer all 20 multiple-choice questions. (2 marks each.) Total 40 marks.

No negative marking.

Only the filled-in boxes will be looked at. No additional commentary or analysis will be read.

Remember to hand up the multiple-choice answer sheet!

1. A problem has a *branching factor* of x . To examine all possible states between the current state and n steps into the future requires examining how many states?
 1. nx
 2. n^x
 3. x^n
 4. x
 5. n
 6. $x + x^2 + x^3 + \dots + x^n$
 7. $x+x^n$
 8. x^x

2. In a perceptron, let inputs be 0 or 1. Let the output be $y = 0$ or 1, and the correct output be C . The Perceptron Learning Rule can be summarised as:
 1. If $y=0$ and $C=1$, then reduce threshold and reduce weights.
 2. If $y=1$ and $C=0$, then reduce threshold and increase weights.
 3. If $y=0$ and $C=1$, then increase threshold and reduce weights.
 4. If $y=1$ and $C=0$, then increase threshold and increase weights.
 5. If $y=0$ and $C=1$, then reduce threshold and increase weights.

3. Neural Networks and GAs can both be seen as hill-climbing on an unknown landscape by taking a finite number of samples. Which of the following is the case:
 1. Neural Networks make a random move, GAs make a random move.
 2. Neural Networks make a directed move, GAs make a directed move.
 3. Neural Networks make a random move, GAs make a directed move.
 4. Neural Networks make a directed move, GAs make a random move.

4. Imagine the following in Best-first search. The heuristic evaluation function tells us a state is 4 steps away from the goal. We examine the children of this state and the heuristic evaluation function tells us all of them are 5 steps away from the goal. How can this happen?
 1. The heuristic evaluation function is inaccurate.
 2. We are in an infinite loop.
 3. This always happens.
 4. We are going the wrong way.
 5. This cannot happen.
 6. The goal has been wrongly defined.

5. One of these is *known* to be untrue:
 1. AI is possible.
 2. In the future, immortal humans will travel to distant stars.
 3. In the future, humans will live forever on earth in artificial bodies, regularly backing up their brains in order to cheat death.
 4. Computers will never be better than humans at chess.
 5. In the future, intelligent machines will exterminate the human race.
 6. AI (the construction of intelligent machines) is impossible.
 7. Computers will never be better than humans at football.

6. Consider the parallelogram whose boundaries are these 4 lines:
 $y=0$, $y=3$, $x=y$ and $(x-3)=y$
 A multi-layer neural network architecture to fire only for points *outside* the parallelogram is:
 1. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an OR gate, then feed the output of that into a NOT gate.
 2. 4 hidden units fire for $y \leq 0$, $y \geq 3$, $x \leq y$, and $(x-3) \geq y$, respectively, then feed their outputs into an AND gate.
 3. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an AND gate.
 4. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an OR gate.
 5. 4 hidden units fire for $y \leq 0$, $y \geq 3$, $x \leq y$, and $(x-3) \geq y$, respectively, then feed their outputs into an AND gate, then feed the output of that into a NOT gate.

6. 4 hidden units fire for $y \geq 0$, $y \leq 3$, $x \geq y$, and $(x-3) \leq y$, respectively, then feed their outputs into an AND gate, then feed the output of that into a NOT gate.

7. The set of inequalities that define a perceptron to implement OR are:
 1. $0 \geq t$, $w_1 < t$, $w_2 < t$
 2. $0 > t$, $w_1 \leq t$, $w_2 \leq t$
 3. $0 \geq t$, $w_1 \leq t$, $w_2 \leq t$
 4. $0 < t$, $w_1 > t$, $w_2 > t$
 5. $0 < t$, $w_1 \geq t$, $w_2 \geq t$
 6. $0 \leq t$, $w_1 > t$, $w_2 > t$
 7. $0 \leq t$, $w_1 \geq t$, $w_2 \geq t$
 8. $0 > t$, $w_1 < t$, $w_2 < t$

8. We are learning to represent a function $Q(x)$. The input x is a 10 dimensional vector. Each dimension is an integer from 1 to 100. One approach to representing this function would be to have a "lookup table" or array of entries, where we have one entry $Q(x)$ for each input x . This array would be initially blank and we would fill it in by learning. The size of the array is:
 1. 10^{100}
 2. 100
 3. 1000
 4. 100^{10}
 5. 10
 6. 1010
 7. 110

9. In a multi-layer neural network, hidden node j can learn to ignore input I_i by:
 1. Moving w_{ji} towards 1.
 2. Moving w_{ji} towards 0.
 3. Moving w_{jj} towards 0.
 4. Moving w_{jj} towards 1.
 5. It cannot learn to ignore input I_i .
 6. Moving w_{ij} towards 1.
 7. Moving w_{ij} towards 0.

10. With large state-spaces, it is hard to draw the entire state-space representation of a problem. This is solved by:
1. Using a heuristic to reduce the size of the state-space.
 2. Using symmetry to reduce the size of the state-space.
 3. Writing a program to automatically draw it.
 4. We draw a partial state-space representation of the problem.
 5. Doing a truncated search.
 6. It is not solved. We do not draw the state-space representation of the problem.
11. Consider the Travelling salesman problem with n cities. We must start in city A and then visit all the other cities. We do not consider paths that visit the same city twice. The number of possible paths is:
1. $n!$
 2. $(n-1)!$
 3. n^2
 4. $2^{(n-1)}$
 5. n^n
 6. 2^n
12. A multi-layer neural network has 1 input node, 12 hidden nodes, and 1 output node. The total number of parameters to be learnt (weights and thresholds) in the network is:
1. 38.
 2. 12.
 3. 13.
 4. 37.
 5. 1.
 6. 24.
 7. 14.
13. We are learning from exemplars. We experience 3 exemplars:
When input is (x_1) output should be (a_1) .
When input is (x_2) output should be (a_1) .
When input is (x_1) output should be (a_2) .
How can this happen?
1. The 3rd exemplar can happen but not the 2nd.
 2. This always happens.
 3. The machine is badly programmed.

4. The 2nd exemplar can happen but not the 3rd.
5. This cannot happen.
6. There are missing inputs, or the world is probabilistic.

14. Given a line in 2-dimensional space:

$$a x + b y = c$$

Define a perceptron that receives x and y as input and fires if and only if the point is on this side of the line:

$$a x + b y \leq c$$

1. weights: a, b , threshold: c
2. weights: a, b , threshold: $-c$
3. weights: $a, -b$, threshold: $-c$
4. weights: $-a, -b$, threshold: $-c$
5. weights: $-a, -b$, threshold: c
6. weights: $-a, b$, threshold: $-c$

15. Human mate choice is a *heuristic* because:

1. Humans do not operate according to logic.
2. Humans cannot compare different solutions in the search space.
3. Humans cannot search the partial search space.
4. Humans are not machines.
5. Humans do not implicitly store all their memories.
6. Humans do not explicitly store all their memories.
7. Humans do not run algorithms.
8. Humans cannot search the entire search space.

16. In machine evolution, let the probability of an individual reproducing be *its fitness divided by the sum of fitnesses of all individuals*. Then:

1. Fitness must be ≤ 0
2. Fitness can have any value.
3. Fitness must be ≥ 0
4. Fitness can have any value except zero.
5. Fitness must be $= 0$
6. Fitness must be > 0

17. Why would we in practice *not* learn the function $y = \sin(x) + \cos(x)$ from exemplars?

1. There is nothing to learn. The problem is solved.
2. We *would* learn it from exemplars.
3. Divide by zero error.
4. Learning it from exemplars is impossible for some values of x .
5. It would take too long.
6. Learning it from exemplars is impossible.
7. It is a chaotic function.

18. A multi-layer neural network has 1 input node, 12 hidden nodes, and 2 output nodes. The total number of parameters to be learnt (weights and thresholds) in the network is:

1. 14.
2. 15.
3. 38.
4. 36.
5. 50.
6. 51.
7. 24.
8. 12.

19. A neural network has 1 input unit, n hidden units, and 1 output unit. The weights on the input layer are all the same: $w_{ij} = W_1$. The thresholds of the hidden units are all the same: $t_j = T$. The weights on the output layer are all the same: $w_{jk} = W_2$.

This network is equivalent to a network with 1 input unit, 1 hidden unit, and 1 output unit, where $(w_{ij}, t_j, w_{jk}) =$

1. $(W_1, n T, W_2)$
2. $(W_1, T, n W_2)$
3. $(n W_1, T, W_2)$
4. (W_1, T, W_2)
5. $(n W_1, T, n W_2)$
6. $(W_1, n T, n W_2)$

20. In trying to predict boxing matches by matching to the nearest exemplar, we have two exemplars:

Opponent height - 186 cm. Opponent weight - 90 kg. Result - I won.

Opponent height - 180 cm. Opponent weight - 95 kg. Result - I lost.

The question is:

Opponent height - 185 cm. Opponent weight - 94 kg. Will I win?

We predict by *Hamming distance* to the nearest exemplar, (a) with height in cm, (b) with height in m. What does it predict?

1. (a) I lose, (b) I win.
2. (a) I win, (b) I win.
3. (a) I lose, (b) I lose.
4. (a) I win, (b) I lose.

Long questions

Answer 2 long questions out of 4. (25 marks each.) Total 50 marks.

- Each question gives **maximum number of sentences** for the answer. Do not exceed this.
- You might like to write your answers as **rough work** first to figure out what sentences to use. Make sure any rough work is crossed out.
- Put your sentences as **bullet points** not a paragraph.
- Do *not* answer more than 2 long questions.
- If you want to scrap one and give me another one, you must cross out the one you are scrapping. I will mark the first 2 (not crossed out) that I see. I will not mark more than 2.

Question 1

[Total marks: 25]

- 1(a) Discuss trying to *maximise* the following function from exemplars. [5 marks]
Explain what would happen during learning.
Input x is a number. It does *not* include the clock time.
The function is defined as follows:

```
if ( x > 10 )  
  f(x) = 1  if clock time is PM.  
  f(x) = -1 if clock time is AM.  
else if ( x < -10 )  
  f(x) = -1  if clock time is PM.  
  f(x) = 1  if clock time is AM.  
else f(x) = 0
```

Maximum length: **3 sentences**

- 1(b) Consider a heuristic search as follows: [5 marks]
 Initialise with 100 random solutions.
 Repeat for many iterations:
 {
 Test the 100 solutions.
 Pick the best n.
 Make small changes in them
 to generate 100 new solutions.
 }
 If n=99 what will this look like?
 Maximum length: **2 sentences**
- 1(c) Mathematicians have long-standing tools to maximise functions and [5 marks]
 calculate function values.
 What is different about the "functions" AI programs try to maximise
 and calculate values for?
 Maximum length: **3 sentences**
- 1(d) We believe that some function exists from n-dimensional input x to [5 marks]
 output f(x). That is, we can set up our system with input x and some
 process will give us a value for f(x). We have no equation for f.
 Contrast *representing* f(x) with *maximising* f(x).
 Maximum length: **3 sentences**
- 1(e) Is this function $y = f(x)$ easy or hard to maximise from *exemplars*? [5 marks]
 $f(1.0372) = 80$.
 $f(1.0373) = 100$.
 $f(x) = 0$ for all other x.
 Maximum length: **3 sentences**

[Total marks: 25]

Question 2

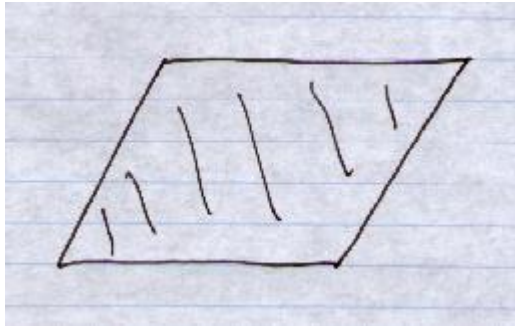
- 2(a) What would Best-first search with a *random* heuristic evaluation [5 marks]
 function look like? (A heuristic evaluation function that generated
 random numbers.)
 Maximum length: **2 sentences**
- 2(b) In the N-puzzle problem, we use a heuristic (distance to goal estimate) [5 marks]
 as follows: $h(n)$ = number of pieces out of place.
 If $h(x) = 1$ then are we close to the goal?
 Maximum length: **4 sentences**
- 2(c) In X's and O's, we considered a heuristic of "Number of winning [5 marks]
 opportunities". You take the next move with the highest number of
 winning opportunities. You are done! No search needed. What is the
 catch?
 Maximum length: **1 sentence**
- 2(d) In a very small search tree, it may not matter what search algorithm we [5 marks]
 use. Why?
 Maximum length: **1 sentence**

- 2(e) Consider using *Algorithm A*. [5 marks]
 The start state has two child states P and Q.
 P has heuristic evaluation (distance from goal) of 3.
 Q has heuristic evaluation 40.
 We look at P's child. It has heuristic evaluation 2.
Its child has heuristic evaluation 2.
Its child has heuristic evaluation 2.
 And so on, forever.
 Show that Algorithm A will deal with this and eventually look at state Q.
 Maximum length: **3 sentences**

Question 3 [Total marks: 25]

- 3(a) The following is the perceptron learning rule. We have multi- [5 marks]
 dimensional input, where each component of the input I_i is binary 0 or 1. It produces output $y = 0$ or 1. We show it the "correct" output $\mathbf{O} = 0$ or 1. We modify the weights and the threshold by the rules:
 $w_i := w_i + C (\mathbf{O} - y) I_i$
 $t := t - C (\mathbf{O} - y)$
 where C is some positive constant.
 Question: What happens on input links with value zero?
 Maximum length: **1 sentence**
- 3(b) In supervised learning with a neural network, we show the network an [5 marks]
 Input, and then show it the correct Output. The network modifies its weights so that next time it sees the input, it will be *more likely* to generate the correct output.
 Why such a roundabout method? Why not simply *remember* the correct output and return it *exactly* next time it is asked?
 Maximum length: **2 sentences**
- 3(c) We are trying to build a character recognition program. Characters are [5 marks]
 images of 100 x 100 pixels. Each pixel is black or white. We are trying to recognise the digits "0" to "9".
 Input is 2 dimensional so therefore the input vector should be 2 dimensional. True or false?
 Maximum length: **2 sentences**
- 3(d) Consider the triangular area formed by the 3 lines: [5 marks]
 $x=0$, $y=0$, $x+y=1$
 Define the weights and thresholds of a *multi-layer* network that will fire only for points within the triangular area.
 Maximum length: **6 sentences**

- 3(e) Explain in English how you can make a multi-layer network that fires if and only if the input point is inside the parallelogram in this figure: [5 marks]



Maximum length: **2 sentences**

Question 4

[Total marks: 25]

- 4(a) In a Neural Network, search can make a *directed* move. Genetic Algorithm search can only make a *random* move. Explain. [5 marks]
Maximum length: **2 sentences**
- 4(b) In a genetic algorithm (GA), let each individual i have fitness $f(i)$, and let its probability of reproducing, $p(i)$, be: [5 marks]

$$p(i) = \frac{e^{\frac{f(i)}{T}}}{\sum_j e^{\frac{f(j)}{T}}}$$

where T is the "temperature".

Show that as $T \rightarrow \infty$ we tend to pick a random individual, no matter what the fitnesses.

Maximum length: **2 sentences**

- 4(c) In a genetic algorithm (GA), let the probability of an individual being picked for reproduction be: [5 marks]

$$p = \frac{\text{my fitness}}{\text{sum of fitnesses of all creatures}}$$

Fitness cannot be zero. Why?

Maximum length: **1 sentence**

- 4(d) In a genetic algorithm: What would happen if rather than picking the fittest, you define a *minimum* fitness that individuals have to reach in [5 marks]

order to reproduce?

Maximum length: **2 sentences**

- 4(e) In a genetic algorithm (GA), let p_m be the probability that any bit in the chromosome mutates at the point where it is inherited from the parent by the child. Let the chromosome be binary, so that mutation means a bit flip. [5 marks]

What would happen if $p_m = 0$ (but you still had crossover)?

Maximum length: **4 sentences**

Bonus question

Bonus question. (10 marks.)

It is "bonus" because I tell you it in advance. The question is to make you do some reading. You can prepare this question in advance.

Bonus question:

Tell me about some AI-related invention *after 2020* that was *not* mentioned on the course.

Answer sheet

Black out the square that matches the correct answer.

Exam No: _____

Seat No: _____

Question	1.	2.	3.	4.	5.	6.	7.	8.
1								
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								